

Lecture Notes on

Unit 02 Differential Equations

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1 Higher Order Linear Differential Equations

1.1 Higher Order Linear ODEs (introduction)

1.2 Higher Order Linear ODEs (continued)

1.3 Second Order Linear ODEs (introduction)

1.4 How to Compute a Wronskian & Linear Independence

2 Higher Order Linear Diff. Eq./ Reduction Of Order

2.1 Reduction of Order (introduction)

2.2 Reduct. of Order (Additional Homogeneous Equations Examples)

2.3 Reduct. of Order (Second-Order Non-Homogeneous Exmaples)

3 Homogeneous Linear Equations with Constant Coefficients

3.1 Second-Order Homogeneous Equations (Constant Coefficients Introduction)

3.2 Second-Order Equations with Constant Coefficients (Distinct Real Roots)

3.3 Second-Order Equations with Constant Coefficients (Repeated Real Roots)

3.4 Second-Order Equations with Constant Coefficients (Complex Roots)

4 Undetermined Coefficients - Superposition Approach

4.1 Higher order linear ODEs

4.2 Nonhomogeneous equations

4.3 Method of Undetermined Coefficients (Second Order Non-Homogeneous Equations)

4.4 Undetermined Coefficients (Multiplying by x for Linear Independence)

5 Undetermined Coefficients - Annihilator Approach

5.1 How to use the Annihilator Method to Solve a Differential Equation

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